

Texuix's Belugomomous zur Steepest Descent

Why Momentum Really Works

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gradient descent: $w^{k+1} = w^k - \alpha \nabla f(w^k)$

momentum: $z^{k+1} = bz^k + \nabla f(w^k)$ and $w^{k+1} = w^k - \alpha z^{k+1}$

Convex quadratic: $f(w) = \frac{1}{2} w^T A w - b^T w$, optimal $w^* = A^{-1}b$ for $\nabla f(w^*) = 0$

$w^{k+1} = w^k - \alpha(Aw^k - b)$ with eigenvectors of A we can view steepest descent with all dimensions independent

$A = Q \text{diag}(\lambda_1, \dots, \lambda_n)^T Q^T$ for $Q = [q_1, \dots, q_n]$ and λ_i sorted

error at step k : $w^k - w^*$, change of basis $x^k = Q^T(w^k - w^*)$

$w^* = A^{-1}b \Rightarrow Aw^* = b \Rightarrow A(w^k - w^*) = Aw^k - b$

and $w^{k+1} = w^k - \alpha(Aw^k - b)$ so $w^{k+1} = w^k - \alpha A(w^k - w^*) \xrightarrow{\text{subtract } w^*}$

$w^{k+1} - w^* = w^k - w^* - \alpha A(w^k - w^*) \Rightarrow$

$w^{k+1} - w^* = (I - \alpha A)(w^k - w^*) \Rightarrow Q^T(w^{k+1} - w^*) = Q^T(I - \alpha A)(w^k - w^*)$

and $I - \alpha A = I - \alpha Q \Lambda Q^T = Q(I - \alpha \Lambda)Q^T$

so $Q^T(I - \alpha A)(w^k - w^*) = Q^T \cdot Q(I - \alpha \Lambda)Q^T(w^k - w^*)$

we have $Q^T \cdot Q = I$, $Q^T(w^k - w^*) = x^k$ and $Q^T(w^{k+1} - w^*) = x^{k+1}$

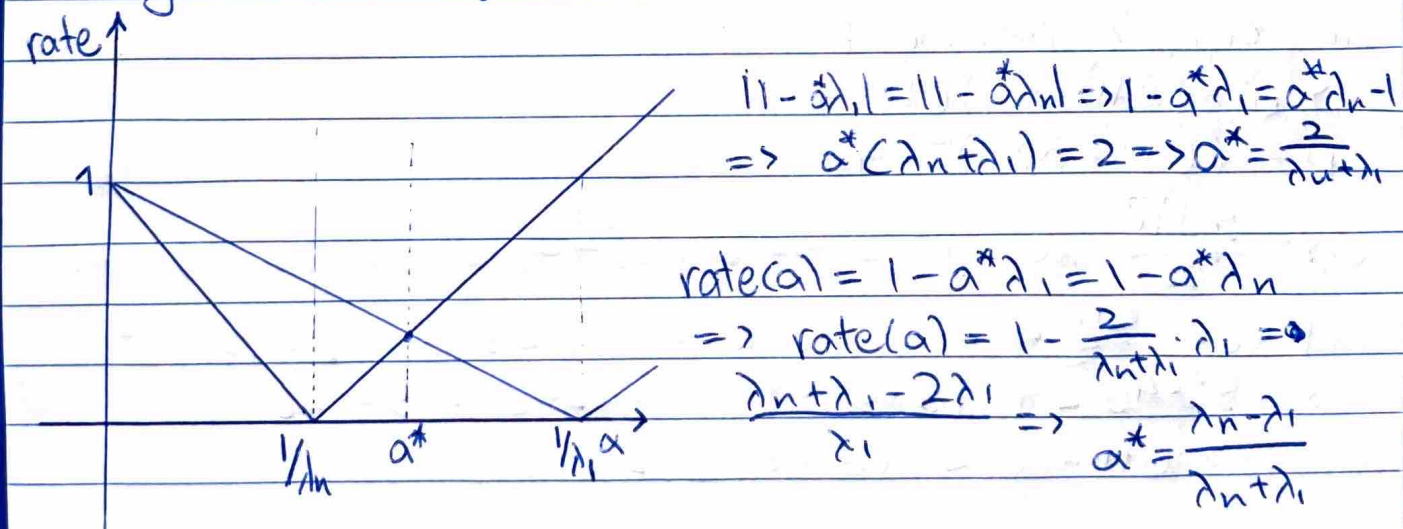
Eventually, $x^{k+1} = (I - \alpha \Lambda)x^k$ and $x_i^{k+1} = (1 - \alpha \lambda_i)^k x_i^k = (1 - \alpha \lambda_i)^k x_i^0$

with $Qx^k = w^k - w^* \Rightarrow w^k - w^* = \sum_{i=1}^n x_i^0 (1 - \alpha \lambda_i)^k \cdot q_i$

where x_i^0 is the component of the initial error $w^0 - w^*$ projected onto the i -th eigenvector q_i , how bad the initial guess was in the direction i and $(1 - \alpha \lambda_i)^k$ is how much the error shrunk after k steps in direction i .

we want $|1 - \alpha \lambda_i| < 1$ to converge $\Rightarrow 0 < \alpha \lambda_i < 2$

convergence rate = $\max_i |1 - \alpha \lambda_i| = \max\{|1 - \alpha \lambda_1|, |1 - \alpha \lambda_n|\}$



therefor, $\alpha^* = \frac{\lambda_1/\lambda_1 - 1}{\lambda_1/\lambda_1 + 1} \xrightarrow{\lambda_1/\lambda_1 \rightarrow \infty} \alpha^* = \frac{k-1}{k+1}$, k : condition number

Now we want to find the rate of momentum cause we have

$$\rho = \frac{k-1}{k+1} \text{ and } \rho^k \leq \epsilon \text{ after } k \text{ steps: } k \geq \frac{\log \epsilon}{\log \rho}$$

$\rho = \frac{k-1}{k+1} = 1 - \frac{2}{k+1} \approx 1 - \frac{2}{k}$ for large k and for small x : $\log(1-x) = -x$ because $\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} \dots$ so $\frac{x^2}{2}, \dots \rightarrow 0 \Rightarrow \log(1-x) = -x$ with Taylor Series

$$\text{Therefor, } \log \rho \approx -\frac{2}{k} \Rightarrow k \geq \frac{\log(1/\epsilon)}{2/k} \Rightarrow k \geq \frac{k}{2} \log\left(\frac{1}{\epsilon}\right) \Rightarrow k = O(k)$$

for $z^{k+1} = bz^k + \nabla f(w^k)$ and $w^{k+1} = w^k - \alpha z^{k+1}$, $\nabla f(w^k) = Aw^k - b$
 $z^{k+1} = bz^k + Aw^k - b$, change of basis with $x^k = Q(w^k - w^*)$, $y^k = Qz^k$

$$z^{k+1} = bz^k + Aw^k - b = bz^k + A(w^k - w^*) \text{ from } Aw^* = b \text{ and } Qz^{k+1} = bQ^T z^k + Q^T A(w^k - w^*)$$

$$\text{and } Q^T A(w^k - w^*) = \Lambda Q^T(w^k - w^*) = \Lambda x^k \text{ and } Qz^{k+1} = y^{k+1}$$

$$\text{so } y^{k+1} = by^k + \Lambda x^k \quad (1)$$

$$\text{from } w^{k+1} = w^k - \alpha z^{k+1} \Rightarrow w^{k+1} - w^* = w^k - w^* - \alpha z^{k+1} \Rightarrow$$

$$Q^T(w^{k+1} - w^*) = Q^T(w^k - w^*) - \alpha Q^T z^{k+1} \Rightarrow$$

$$x^{k+1} = x^k - \alpha(by^k + \Lambda x^k) = (I - \Lambda)x^k - \alpha by^k \quad (2)$$

$$\xrightarrow{(1), (2)} \begin{pmatrix} y_i^{k+1} \\ x_i^{k+1} \end{pmatrix} = \begin{pmatrix} b & \lambda_i \\ -\alpha b & 1 - \alpha \lambda_i \end{pmatrix} \begin{pmatrix} y_i^k \\ x_i^k \end{pmatrix} = R^{k+1} \begin{pmatrix} y_i^0 \\ x_i^0 \end{pmatrix} \text{ for } R = \begin{pmatrix} b & \lambda_i \\ -\alpha b & 1 - \alpha \lambda_i \end{pmatrix}$$

now ϕ_1, ϕ_2 eigenvalues of R and characteristic polynomial $|R - \phi I| = 0$
 $\Rightarrow \phi^2 - (b+1-\alpha\lambda_i)\phi + b = 0$

$$D = (b+1-\alpha\lambda_i)^2 - 4b, S = (b+1-\alpha\lambda_i) \text{ so } D = S^2 - 4b$$

$$\phi_{1,2} = \frac{S \pm \sqrt{D}}{2} \text{ we want } \phi_1 \rightarrow 0 \text{ and } \phi_2 \rightarrow 0 \text{ so } \max\{|\phi_1|, |\phi_2|\} < 1$$

$$\text{first case: } \frac{S + \sqrt{D}}{2} < 1 \Rightarrow S + \sqrt{D} < 2 \Rightarrow \sqrt{D} < 2 - S \Rightarrow D < (2 - S)^2$$

$$\Rightarrow S^2 - 4b < 4 - 4S + S^2 \Rightarrow 4S - 4 < 4b \Rightarrow S - 1 < b$$

$$\Rightarrow b+1-\alpha\lambda_i < 1+b \Rightarrow \alpha\lambda_i > 0 \quad (3)$$

Heavy Ball method (Polyak 1964)

second case: ~~$S^2 + 4S + 4 > S^2 - 4\theta$~~ $\frac{S - \sqrt{D}}{2}, -1 \Rightarrow S - \sqrt{D} > -2$
 $\Rightarrow S + 2 > \sqrt{D} \Rightarrow (S + 2)^2 > D \Rightarrow$
 $S^2 + 4S + 4 > S^2 - 4\theta \Rightarrow S + 1 + \theta > 0 \Rightarrow \theta + 1 - \alpha\lambda_i + \theta + 1 > 0 \Rightarrow$
 $2\theta - \alpha\lambda_i + 2 > 0 \Rightarrow \alpha\lambda_i < 2 + 2\theta \quad (4)$

(3), (4) $\Rightarrow 0 < \alpha\lambda_i < 2 + 2\theta$ with $\theta < 1$

for $D > 0$: increasing θ means D shrinks and for $D < 0$: decreasing θ also shrinks D so we find the optimum for $D = 0$

solve for $D = 0 \Rightarrow S^2 = 4\theta \Rightarrow S = 2\sqrt{\theta}$, $\rho = \frac{S}{2} = \frac{2\sqrt{\theta}}{2} = \sqrt{\theta} \leftarrow \text{rate}$
 and $S = 2\sqrt{\theta} \Rightarrow \theta + 1 - \alpha\lambda_i = 2\sqrt{\theta} \Rightarrow (\sqrt{\theta} - 1)^2 = \alpha\lambda_i \Rightarrow$
 $\sqrt{\theta} = 1 - \sqrt{\alpha\lambda_i}$ and we have rate $\rho = \sqrt{\theta} = 1 - \sqrt{\alpha\lambda_i}$

max rate = $\max \{1 - \sqrt{\alpha\lambda_1}, 1 - \sqrt{\alpha\lambda_n}\}$ find α^* optimum for
 $1 - \sqrt{\alpha^*\lambda_1} = \sqrt{\alpha^*\lambda_n} - 1 \Rightarrow 2 = \sqrt{\alpha^*\lambda_n} + \sqrt{\alpha^*\lambda_1} \Rightarrow \sqrt{\alpha^*} = 2 / (\sqrt{\lambda_n} + \sqrt{\lambda_1})$

$\Rightarrow \alpha^* = \frac{4}{(\sqrt{\lambda_1} + \sqrt{\lambda_n})^2}$ and $\theta^* = (1 - \sqrt{\alpha^*\lambda_1})^2 = \left(1 - 2 \frac{\sqrt{\lambda_1}}{\sqrt{\lambda_n} + \sqrt{\lambda_1}}\right)^2$

$\Rightarrow \theta^* = \left(\frac{\sqrt{\lambda_n} - \sqrt{\lambda_1}}{\sqrt{\lambda_n} + \sqrt{\lambda_1}}\right)^2$ and $\alpha^* = \frac{4}{(\sqrt{\lambda_n} + \sqrt{\lambda_1})^2}$

$\rho = \sqrt{\theta^*} = \frac{\sqrt{\lambda_n} - \sqrt{\lambda_1}}{\sqrt{\lambda_n} + \sqrt{\lambda_1}} = \frac{\sqrt{\lambda_n/\lambda_1} - 1}{\sqrt{\lambda_n/\lambda_1} + 1} \xrightarrow{\kappa = \lambda_n/\lambda_1} \rho = \text{optimal rate} = \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}$

steepest descent: $O(\kappa)$, with momentum: $O(\sqrt{\kappa})$

for the "Linear First Order Methods" class: $w^{k+1} = w^0 + \sum_{i=1}^k \Gamma_i^k \nabla f(w^i)$ $\rightarrow$ diagonal

and using the convex Rosenbrock: $f(w) = \frac{1}{2}(w_1 - 1)^2 + \frac{1}{2} \sum_{i=2}^n (w_i - w_{i+1})^2 + \frac{1}{2} \|w\|^2$
 the optimal solution is $w_i^* = \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}\right)^2$ so Nesterov proved that
 no algorithm in the class above can do better than this so
 in some ways steepest descent with momentum is optimal for
 quadratic functions

NAG: Nesterov Accelerated Gradient (1983)

An overview of gradient descent optimization algorithms) reader.io

small $\alpha_k \rightarrow$ slow convergence, α_k : learning rate

large $\alpha_k \rightarrow$ loss function fluctuates

momentum: $z^{k+1} = \beta z^k + \nabla f(w^k)$, $w^{k+1} = w^k - \alpha z^{k+1}$, blindly follow slope

Nesterov Accelerated Gradient (NAG): know where the ball is going

$y^k = w^k + \beta(w^k - w^{k-1})$, y^k : lookahead point, jump β times the last displacement, y^k is where momentum alone would take us

$w^{k+1} = w^k - \alpha \nabla f(y^k)$, $\nabla f(y^k)$: the slope of where we are going

$\alpha \nabla f(y^k)$: correction step

momentum has rate $\frac{k-1}{k+1}$ on quadratics: $f(x) = \frac{1}{2} x^T Q x - b^T x + c$, $x \in \mathbb{R}^n$

NAG has rate $\frac{k-1}{k+1}$ for all convex functions: $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$

for $\lambda \in [0,1]$ and points x, y of $f \rightarrow$ graph looks U-shape

the type suggests that the graph of the function lies below the straight line connecting any two points

sparse data $\rightarrow$ adaptive learning-rate methods, α_k : learning-rate, step-size

$\alpha^* = 4/(\lambda_{\min} + \lambda_{\max})^2$, $\beta^* = 1/k$, rate $= \frac{k-1}{k+1}$ for NAG on convex functions

Two-Point Step Size Gradient Methods - J. Barzilai and J. Borwein)

for "one-point" step-size we only use g_k to determine α_k and nothing from the past where $g_k = \nabla f(x_k)$

for $\frac{1}{2} x^T Q x - b^T x$, we have $O(n^2)$ complexity: $\alpha_k = \frac{g_k^T g_k}{g_k^T Q g_k}$, for $Q g_k$

BB: $x_{k+1} = x_k - S_k g_k$, where $g_k = \nabla f(x_k)$, $S_k = \alpha_k I$

so $x_{k+1} = x_k - \alpha_k g_k$, same direction always but we want α_k that minimizes $\| \Delta x - \alpha \Delta g \|^2$, $\Delta x = x_k - x_{k-1}$, $\Delta g = g_k - g_{k-1}$

so we have $\alpha_k = \frac{\langle \Delta x, \Delta g \rangle}{\langle \Delta g, \Delta g \rangle}$, $\langle a, b \rangle$: scalar product $O(n)$, (inner product)

this is "the short-step" - BB2 for minimizing $\| \Delta x - \alpha \Delta g \|^2$

"the long-step" - BB1 for minimizing $\| \alpha \Delta x - \Delta g \|^2$ we have $\alpha_k = \frac{\langle \Delta x, \Delta x \rangle}{\langle \Delta x, \Delta g \rangle}$

$\langle a, b \rangle = \sum_{i=1}^n a_i b_i = (a_1 b_1 + \dots + a_n b_n)$

Κεφάλαιο 3 - Μέθοδοι Αυξίμου Εύθείας)

Αλγόριθμος Steepest Descent:

- 1) Αρχική προβλεπτή διάνυσμα x_0 και σφάλμα $\epsilon > 0, k=0$
- 2) α) Αν $\|\nabla f(k)\| < \epsilon$, τότε τερματίζει και προσεγγιστική τιμή $x_k = x_k$
β) αλλιώς, $p_k = -\nabla f(x_k)$
γ) βρίσκουμε α_k που ελαχιστοποιεί την $\varphi(\alpha_k) = f(x_k - \alpha_k \nabla f(x_k))$
ως προς $\alpha_k > 0$
δ) θέτουμε $x_{k+1} = x_k - \alpha_k \nabla f(x_k)$
ε) επαναλαμβάνουμε από (α) για $k=k+1$

Η πορεία προς το ελάχιστο είναι μία τετραγωνική γραμμή με μεγάλο αριθμό διαδοχικών εγγεγραμμένων τμημάτων τα οποία σχηματίζουν γωνία 90° μεταξύ τους. Άρα η μέθοδος είναι αργή και αναποτελεσματική. Αν x_k με x^* διαφορά εγγραμμένο του $\nabla f(x_k)$, ο αλγόριθμος τερματίζει βε ένα βήμα.

Για $f(x) = \frac{1}{2}x^T Q x - b^T x + c$, $x, b \in \mathbb{R}^n$, Q συμμετρικός $n \times n$, $c \in \mathbb{R}$
Έχουμε, $\nabla f(x) = Qx - b$, $\nabla^2 f(x) = Q$, $x^* = Q^{-1}b$
και η λύση του προβλήματος $\min_{x \in \mathbb{R}^n} f(x)$ είναι $a = -\frac{p^T \nabla f(x)}{p^T Q p}$

Κεφάλαιο 4 - Η Μέθοδος Newton Στην Αδέσμευτη Βελτιστοποίηση)
η Steepest Descent στην τετραγωνική εγγραμμένη δίνει στην εκτίμηση του σφάλματος: $\|x_{k+1} - x^*\|_Q^2 \leq \left(\frac{\lambda_n - \lambda_1}{\lambda_n + \lambda_1}\right)^2 \|x_k - x^*\|_Q^2$
για $\lambda_1 \leq \dots \leq \lambda_n$ ιδιοτιμές του Q , άρα συγκλίνει γραμμικά